**Rock Type Classification** | *Machine learning | Ensemble Learning | Python* **Sept 2024 – Oct 2024**

- Implemented and optimized **Softmax Regression, SVM (82.2%), and Random Forest (76.7%)** models for classifying rock types, leveraging features like texture and stripes. Developed and compared ensemble models **(Hard Voting: 81.1%, Soft Voting: 77.8%, Stacked: 78.9%)** against human classification accuracy to evaluate reliability and performance.

**Hindi Sentiment Analysis on Tweets** | *Naive Bayes | Lexicon-Based Model | NLP, Python* **Oct 2023 – Jan 2024**

- Increased accuracy of Hindi sentiment analysis from **41% to 65%** by enhancing the **lexicon-based method**, addressing a key research gap, and improving sentiment analysis for Hindi tweets.

**PLANTIFY - Mobile Application** | *Image Segmentation | CNN, Python* **Jan 2020 – May 2020**

- Achieved 70% accuracy in plant recommendation by leveraging a **U-Net model for image segmentation**, effectively identifying sky-ground boundaries, and providing tailored plant recommendations based on location and solar capabilities.

**Rock Type Classification** | *Machine learning, Ensemble Learning* **Sept 2024 – Oct 2024**

- Implemented and optimized **Softmax Regression, SVM (82.2%), and Random Forest (76.7%)** models for classifying rock types, leveraging features like texture and stripes. Developed and compared ensemble models **(Hard Voting: 81.1%, Soft Voting: 77.8%, Stacked: 78.9%)** against human classification accuracy to evaluate reliability and performance.

**Prostate Cancer Detection** | *Python | Computer Vision| Pytorch| CNN| U-Net* **Nov 2025 - Dec 2025**

- Built a **U-Net–based prostate MRI segmentation** pipeline using MONAI, comparing **zero-shot inference** with **fine-tuning** on medical imaging data; trained and evaluated on **IU BigRed200**, achieving a significant Dice score improvement **(0.415 to 0.774).**

**AI for Early Kidney Disease Detection** | *Snowflake| Stream-lit | ML, Python* **Feb 2025 – May 2025**

- Led the project to perform ***data preprocessing, feature selection, data modeling,*** and ***data visualization***, building scalable pipelines, dashboards, and developed and **deployed ML models** for early kidney disease detection.

**Rock Type Classification** | *Machine learning | Ensemble Learning | Python* **Sept 2024 – Oct 2024**

- Implemented and optimized **Softmax Regression, SVM (82.2%), and Random Forest (76.7%)** models for classifying rock types, leveraging features like texture and stripes. Developed and compared ensemble models **(Hard Voting: 81.1%, Soft Voting: 77.8%, Stacked: 78.9%)** against human classification accuracy to evaluate reliability and performance.

**PLANTIFY - Mobile Application** | *Image Segmentation| Computer Vision | CNN, Python* **Jan 2020 – May 2020**

- Achieved 70% accuracy in plant recommendation by leveraging a **U-Net model for image segmentation**, effectively identifying sky-ground boundaries, and providing tailored plant recommendations based on location and solar capabilities.

**Multimodal Conversational AI System (Text + Tools) | Python, LangChain, Streaming APIs**

- Built **real-time conversational agent integrating text input, tool execution**, and dynamic context updates
- Simulated streaming interaction pipeline (incremental responses, partial context updates) to mimic real-time conversational behavior
- Optimized system for **latency, response coherence, and context tracking across multi-turn interactions**

**Movie Recommendation System** |*Collaborative Filtering | Recommendation System| SVD| Python* **Apr 2021 – Jun 2021**

- Developed a collaborative filtering-based movie recommendation system using matrix factorization techniques, such as Singular Value Decomposition (SVD.
- Implemented algorithms to predict user preferences by analyzing user-item interaction data, achieving high accuracy in recommending movies tailored to individual user profiles.
- Enhanced recommendation accuracy by applying advanced techniques in collaborative filtering, improving the system's predictive performance on unseen data.

**PLANTIFY - Mobile Application** | *Image Segmentation | CNN, Python* **Jan 2020 - May 2020**

- Generated plant recommendations by interpreting data for a user's location and solar capabilities.
- Utilized a U-Net model for image segmentation to identify the boundary between the sky and ground and then encoded the results in a run-length format with an ***accuracy of 70%***.

**Mental Health prediction of Instagram users** | *Machine Learning and RNN | Python* **Aug 2019 - Nov 2019**

- Leveraged deep learning and machine learning methods to forecast the mental health status of Instagram users.
- Performed multi-class and multi-label classification on Instagram posts.

## LEADERSHIP & COMMUNITY INVOLVEMENT

**Author, Midwest Speech and Language Days (MSLD 2025)** **Attendee, SWE (Society of Women Engineers) Conference (2024)**

**Member, SWE (Society of Women Engineers) Conference (2025 & 2024)** **Member, Grace Hopper Celebration (2025)**

## PUBLICATIONS AND PRESENTATIONS

- **Bajpai A.,** D. Cavar, et al, Improving LLM Reasoning Through Ontology-driven Knowledge Graphs: A Comparative Study of Generating Ontologies for Medical RAGs, Midwest Speech and Language Days 2025, Indiana University Bloomington, April 2025.
- **Bajpai A.**, Garg M., Hindi Sentiment Analysis on Tweets, 2nd International Conference on Artificial Intelligence: Theory and Applications (AITA 2024). [SCOPUS][DBLP]
- Garg M., **Bajpai A.**, Kumari S., Rathi S., Dahiya T., Ghosh A., AI-Driven Radiomics for Early Detection of Gynecological Cancers: A Multimodal Approach, International Conference on Smart Cyber Physical Systems (ICSCPS-2024). [SCOPUS] - **Accepted & Presented**

## SKILLS

**Programming Language:** Python, R, SQL, C/C++, PostgreSQL, BigQuery, Linux, Unix

**Frameworks/Tools**: Jenkins, Git, GCP, DevOps, Docker, Kubernetes, PySpark, LangChain, RAG, Azure, AWS, MLflow, ApacheA Airflow, Agile

**AI/ML:** PyTorch, Natural Language Processing (NLP), Neural Networks (including CNN, RNN, LSTM, GRU, FeedForward Networks), Deep Learning, Generative AI(Gen AI), Recommendation System, MLOps, Anomaly Detection, LLMs, Graph NN, Vertex AI, Prompt Engineering

**Libraries/Software:** Numpy, Matplotlib, Pandas, NLTK, scikit-learn, OpenCV, Grafana, Snowflake, Streamlit, FastAPI, Keras, TensorFlow

**Core Computer Science**:Data Structures, Algorithms, Distributed Computing, System Design, Transformer Models, Reinforcement learning

## SKILLS

**Programming Language:** Python, R, SQL, C/C++, PostgreSQL, BigQuery, Linux, Unix, etl

**Frameworks/Tools**: Jenkins, Git, GCP, DevOps, Docker, Kubernetes, PySpark, LangChain, RAG, Azure, AWS, MLflow, Apache Spark, Airflow, Agile

**AI/ML:** PyTorch, Natural Language Processing (NLP), Neural Networks (including CNN, RNN, LSTM, GRU, FeedForward Networks), Deep Learning, Generative AI(Gen AI), Recommendation System, MLOps, Anomaly Detection, LLMs, Graph NN, Vertex AI, Prompt Engineering

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**Core Computer Science**:Data Structures, Algorithms, Distributed Computing, System Design, Transformer Models, Reinforcement learning

**Senior Software Engineer, Optum (UnitedHealth Group)** **Jun 2021 – Jun 2024**

Coverage Feed

- Reduced time from ***500+ minutes to less than 3 minutes*** *by* refactoring existing legacy systems with high code quality standards.
- Optimized error report generation system, reducing errors from ***25,000 to 20***, improving latency, and minimizing production costs through **automated testing** and **CI/CD** integration.

Insurance Letter Generation

- Took initiatives to improve code quality & engineering principles by introducing design patterns, writing unit tests with *at least 90% code coverage*, creating code review guidelines, and mentoring juniors across teams with a growth mindset.
- Built a scalable system to send Compliance Notices as part of ***"The Paperless Initiative"***, saving the company ***$500,000***

***annually***
through **analytical process improvements**.

## VOLUNTEERING EXPERIENCE

**National Service Scheme - Education and Cultural Team Head** **Aug 2018 – Jul 2020**

- Organized and managed blood donation camps, marathons, and plantation drives.
- Led educational initiatives, designed curricula, and participated in community projects, including food distribution and cleanliness drives.

**Saksham: Child Welfare & Educational Trust** - **Volunteer** **Dec 2019 - Feb 2020**

- Actively participated in initiatives promoting child welfare and education, contributing to the organization's overarching goals.

**Vegan Outreach - Volunteer** **Feb 2019 - Mar 2019**

- Engaged in vegan outreach, promoting plant-based living and animal welfare. Collaborated on events and campaigns to raise awareness.

## RESEARCH INTERESTS

Machine Learning, Data Science, Data Mining, Artificial Intelligence,Neural Networks (CNN, LSTM, RNN), Generative AI, Natural Language Processing, Computer Vision, Recommendation Systems

Extra projects
https://github.com/KonuTech/mlops-zoomcamp-project
https://github.com/expectbugs/agents